AXH Series

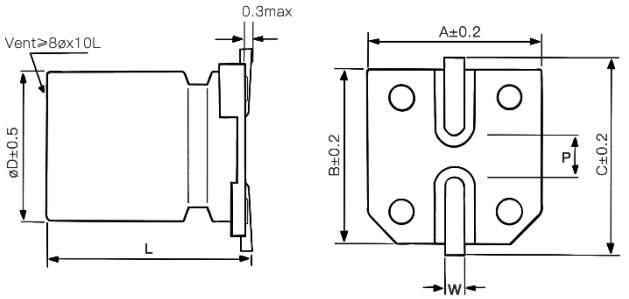
Features

- ϕ8 ~ ϕ12.5, 135℃, 2000 hours assured
- Low impedance capacitors
- For automobile modules and other high temperature applications
- Designed for reflow soldering
- Designed for surface mounting on high-density PCB
- Vibration resistant structure
- RoHS compliant, REACH&SVHC compliant
- AEC-Q200 compliant, Please contact Jarson for more details, test data, information

Marking color: Black

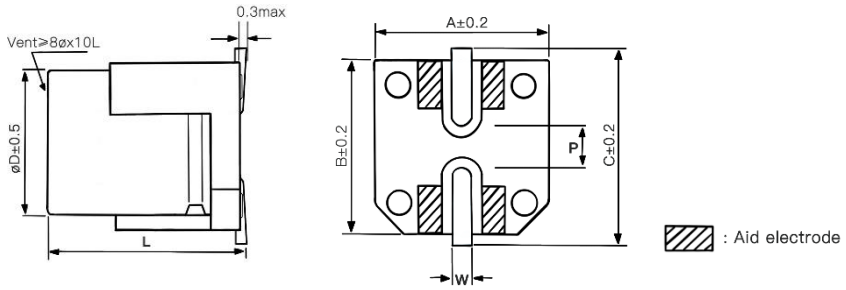
Specifications							
Category temp. range	–40℃ to +135℃						
Capacitance tolerance	±20% (120 Hz / +20 ℃)						
Leakage current	I ≦ 0.01 CV or 3 μA whichever is greater (after 2 minutes)						
Tan δ	Please see the attached characteristics list						
Characteristics at low temperature	Rated voltage (V)	10	16	25	35	50	Impedance ratio at 120 Hz
	Z (-40 ℃) / Z (+20 ℃)	12	8	6	4	4	
Endurance	After applying rated working voltage for 2000 hours at +135 ℃ ± 2 ℃, and then being stabilized at +20 ℃, capacitors shall meet the following limits.						
	Capacitance change	Within ±30% of the initial value					
	Dissipation factor (tan δ)	Less than 300% of the initial value					
	Leakage current	Within the initial limit					
Shelf life	After storage for 1000 h at +135 ℃ ± 2 ℃ with no voltage applied and then being stabilized at +20 ℃, capacitors shall meet the limits specified in endurance.						
Resistance to soldering heat	After reflow soldering and then being stabilized at +20 ℃, capacitors shall meet the following limits.						
	Capacitance change	Within ±10% of the initial value					
	Dissipation factor (tan δ)	Within the initial limit					
	Leakage current	Within the initial limit					
Ripple Current and Frequency Multipliers	Frequency	50Hz	120Hz	300Hz	1kHz	10kHz≦	
	Multiplier	0.35	0.5	0.64	0.83	1.0	

Dimensions:



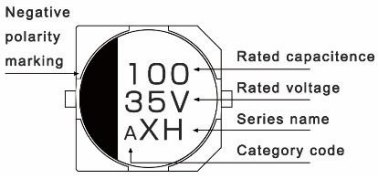
Dimensions						Unit: mm
ϕD	L	A	B	C	W	P±0.2
8	10.5±0.5	8.3	8.3	9.1	0.7~1.3	3.1
10	10.5±0.5	10.3	10.3	11.1	0.7~1.3	4.4
12.5	13.5±0.5	13.0	13.0	14.0	1.1~1.4	4.4

Vibration resistant structure:

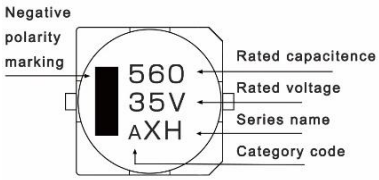


Marking:

ϕD ≦ 10mm



ϕD ≧ 12.5mm



Characteristics list

Rated voltage (V)	Capacitance (±20%) (μF)	Case size		Specification			Part Number④	Taping&Reel
		øD (mm)	L (mm)	Rated ripple current① (mA rms)	Imp.② (Ω)	tan δ③		MPQ (pcs/reel)
10	220	8	10.5	270	0.20	0.30	AXH1A221M0810	500
	330	8	10.5	270	0.20	0.30	AXH1A331M0810	500
		10	10.5	500	0.15	0.30	AXH1A331M1010	500
	470	10	10.5	500	0.15	0.30	AXH1A471M1010	500
16	100	8	10.5	270	0.20	0.23	AXH1C101M0810	500
	220	8	10.5	270	0.20	0.23	AXH1C221M0810	500
	330	10	10.5	500	0.15	0.23	AXH1C331M1010	500
	470	10	10.5	500	0.15	0.23	AXH1C471M1010	500
25	100	8	10.5	270	0.20	0.18	AXH1E101M0810	500
	220	10	10.5	500	0.15	0.18	AXH1E221M1010	500
	330	10	10.5	500	0.15	0.18	AXH1E331M1010	500
	470	12.5	13.5	750	0.08	0.18	AXH1E471M1313	200
	820	12.5	13.5	750	0.08	0.18	AXH1E821M1313	200
	1000	12.5	13.5	750	0.08	0.18	AXH1E102M1313	200
35	47	8	10.5	270	0.20	0.16	AXH1V470M0810	500
	100	8	10.5	270	0.20	0.16	AXH1V101M0810	500
	220	10	10.5	500	0.15	0.16	AXH1V221M1010	500
	330	12.5	13.5	750	0.08	0.16	AXH1V331M1313	200
	470	12.5	13.5	750	0.08	0.16	AXH1V471M1313	200
	560	12.5	13.5	750	0.08	0.16	AXH1V561M1313	200
50	47	8	10.5	270	0.30	0.16	AXH1H470M0810	500
	100	10	10.5	500	0.25	0.16	AXH1H101M1010	500
	220	12.5	13.5	750	0.18	0.16	AXH1H221M1313	200

① Rated ripple current (100kHz / +135℃) ② Impedance (100kHz / +20℃) ③ tan δ (120Hz / +20℃)

※Please refer to the page of reflow conditions for reflow profile.

Part Number System:

SMD E-Cap	AXH series	16V	220μF	±20 %	8ϕx10.5L	
A	**XH**	**1C**	**221**	**M**	**0810**	
Category	Series name	Rated voltage	Capacitance	Tolerance	Case Size	Supplement

④ Note: For more details, please refer to "Part Number System - SMD Type"